

Innovative Integration of Renewable energy systems and advanced materials for Sustainable Seawater Desalination

Abstract

This paper presents a comprehensive analysis of the innovative integration of renewable energy systems and advanced materials for sustainable seawater desalination. Faced with global water scarcity and the significant energy consumption and environmental drawbacks of conventional desalination, this work explores holistic solutions aimed at enhancing efficiency, sustainability, and environmental compatibility. The study systematically evaluates the application of diverse renewable energy sources including geothermal for baseload power, solar for thermal and photovoltaic driven processes, and wind for mechanical and electrical energy to power desalination plants. Methodologically, it examines breakthroughs in material science, focusing on nanostructured materials like carbon nanotubes (CNTs) and graphene oxide (GO) that promise higher flux and salt rejection, alongside biomimetic membranes and advanced antifouling coatings that improve operational longevity. A central theme is the exploration of the water-energy nexus through hybrid systems that mitigate energy intermittency and ensure operational stability.

Key findings from a comparative life cycle assessment (LCA) reveal that renewable powered desalination can reduce greenhouse gas emissions by over 90% compared to fossil fuel-based counterparts. While capital costs are higher, techno economic analyses demonstrate a clear trajectory towards cost-competitiveness, driven by long-term operational savings and technological advancements. The significance of these integrated systems is further illustrated through their application in specific sectors, including sustainable irrigation in agriculture, water recycling, ensuring water security for island and off-grid communities, and providing rapid-response water supplies in disaster-stricken areas. Looking forward, the paper outlines future roadmaps centered on pioneering technologies such as the valorization of desalination brine into salt-based Phase Change Materials (PCMs) for thermal energy storage and the development of low-energy, renewable-coupled electrochemical processes like Capacitive Deionization (CDI). This integrated approach signifies a paradigm shift towards achieving net-zero energy desalination and facilitating a circular economy for brine management, aligning water production with global sustainability targets.

Keywords : Seawater Desalination, Renewable Energy Integration, Advanced Materials, Water-Energy Nexus, Life Cycle Assessment (LCA), Capacitive Deionization (CDI)

Introduction

The rapid pace at which human activity, including but not limited to population growth, industrialization, and climate change, is straining not only global freshwater shortages but also the geopolitical significance of clean water to an alarming level. 1. Potable water has

gone from a basic survival imperative to a geopolitical resource [1,2]. Desalination, the process of extracting freshwater from brackish or seawater, has emerged as a frontier in technology to mitigate against this issue. As of the year 2025, desalinated water accounts for a significant proportion of potable water in desert, semi-arid, and arid regions, namely in the economically prosperous nations of the Middle East, as seen in Fig.1, where the percentages of dependence usually fall above 50% of drinking water total consumption [3,107]. While desalination is a sustainable means of addressing the world's water shortage problem, it is linked with high energy costs, operational costs, and adverse environmental impacts, especially in relation to greenhouse gas emissions and effective brine disposal management [4,5].

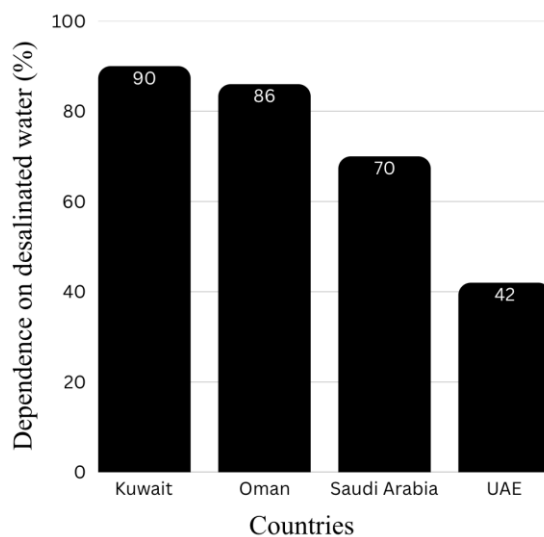


Fig. 1 Graph showing dependence of various countries on desalinated water

This study explores holistic solutions with innovative materials and renewable energy to enhance the efficiency and sustainability of seawater desalination. Our study comprises six connected sections: Section 1 covers different renewable energy sources—solar, wind, and geothermal—utilized for driving desalination plants, such as a comparative study of scalability, intermittency, and integrability feasibility [6,7,8]. Section 2 covers advances in molecular design and material science for adsorbents and membranes, such as fouling prevention, ion selectivity, and increased lifespan [9,10]. Section 3 examines the integration of water and energy using hybridization techniques that amalgamate various renewable resources to guarantee uninterrupted and enhanced plant functionality [8]. Section 4 presents an exhaustive environmental and economic life-cycle analysis comparing renewable-supplied desalination processes with conventionally fossil fuel-based processes [3]. Section 5 covers

areas of application for agriculture, wastewater reuse, island communities, and disaster relief, respectively, along with corresponding case studies [11]. Section 6 anticipates innovative concepts related to thermal storage through phase change materials (PCM) and electro-driven desalination processes and outlines possible research directions [12].

By integrating these parameters, the study makes a technical and strategic input that drives desalination towards net-zero energy operation and facilitates circular management of the brine, while making water production compatible with international sustainability targets [4,13].

1. Different Types of Renewable Energy Systems for Desalination

1.1 Geothermal Energy Systems

Geothermal desalination utilizes the Earth's internal heat—no wind ballads, no solar azimuths—to economically drive desalination, especially where there are assured heat sources [14,15]. Iceland, Japan, the Philippines, New Zealand, Turkey, Mexico, and the US dominate world geothermal capacity and consequently are natural geothermal desalination plant opportunities [16]. Geothermal desalination offers a weather-independent, round-the-clock operation—compared to solar or wind—and so is highly reliable [16]. Two general application approaches exist direct thermal driving and indirect power generation.

1.1.1 Direct Thermal Methods

Multi-Stage Flash (MSF) and Multi-Effect Distillation (MED) remain in widespread use because they can be used at high temperatures (over 70 °C), particularly in cogeneration schemes where waste heat is readily available [17]. MSF today produces about 26% of the world's desalinated water production, even though Reverse Osmosis (RO) has replaced it in new plants [18]. Access to geothermal resources means direct supply of low-carbon heat to MSF/MED plants. Sudden vaporization, as the name suggests, the process includes the distillation of the seawater to steam in a multi-stage flashing process. Seawater is fed into a pressure chamber at low pressure, and through pressure reduction, it evaporates immediately. After a cooling process, the vapor delivers desalinated water. The process is responsible for about 60% of the world's production of desalinated water, as quoted by different sources. Additionally, the thermal energy taken by the brine heater in the form of steam is usually derived from an industry near the desalination plant [19]. Whereas, in Multi-stage flash, seawater is re-heated to 90-120 °C in a multi-stage low-pressure steam brine heater and flashed in multiple stages to evaporate it for the generation of desalinated water. The greatest merit of MSF is the increased tolerance for contaminant loading, i.e., heavy metals, oil, and grease in the feed seawater [20]. However, in the Geothermal ejectors method, the basis of

the ejector mechanisms is high-pressure geothermal steam, which is employed to power a high-pressure jet through a nozzle. It is a pressure-to-kinetic energy conversion process that enables the generation of a vacuum within the evaporation chamber and subsequently improves the generation of freshwater for consumption [21].

1.1.2 Indirect conversion through energy

Geothermal power can be used to drive an Organic Rankine Cycle (ORC) turbine to produce electricity to drive more efficient membrane-based desalination methods, such as Reverse Osmosis (RO) as depicted in Fig 2, and Electrodialysis (ED), and also facilitate system flexibility and hybrid configuration compatibility [22]. The Organic Rankine Cycle (ORC) is a thermodynamic cycle used for energy generation from low to medium, semi-high-grade thermal sources in an 80 to 400 °C temperature range. Geothermal energy here serves as the motive for an ORC engine to generate electrical power. The generated electricity is then utilized to drive reverse osmosis (RO) or electrodialysis (ED) processes for desalination. RO and ED both entail the extraction of impurities by a semipermeable membrane under applied pressure, whereas in ED, an electric field is utilized to attain the same condition [23]. The Kalina cycle, on the other hand, is a Rankine cycle with the only difference being that the Kalina Cycle uses a solution of two substances as the working fluid—water and ammonia. The cycle remains the same; the geofluid of the cycle is 120 °C and 2 bar pressure, which produces 932.5 kW of power. Brine is used for a 4-stage desalination system [24]. Hybrid Systems, as the name implies, a blend of two geothermal power generation approaches, e.g., RO and MD, can maximize energy obtained and reduce heat loss in the form of waste [25]. Benefits are minimal fossil fuel dependence, high capacity factor, and low carbon footprint [16,21]. Geothermal desalinated water is priced around \$1.50 per cubic meter, as estimated by the 2011 US Department of Energy, and is as economical as conventional sources [16]. Limitations are dependence on geographical limitations (geothermal wells are location-specific) and high initial investment in drilling and thermal plant integration [21,22].

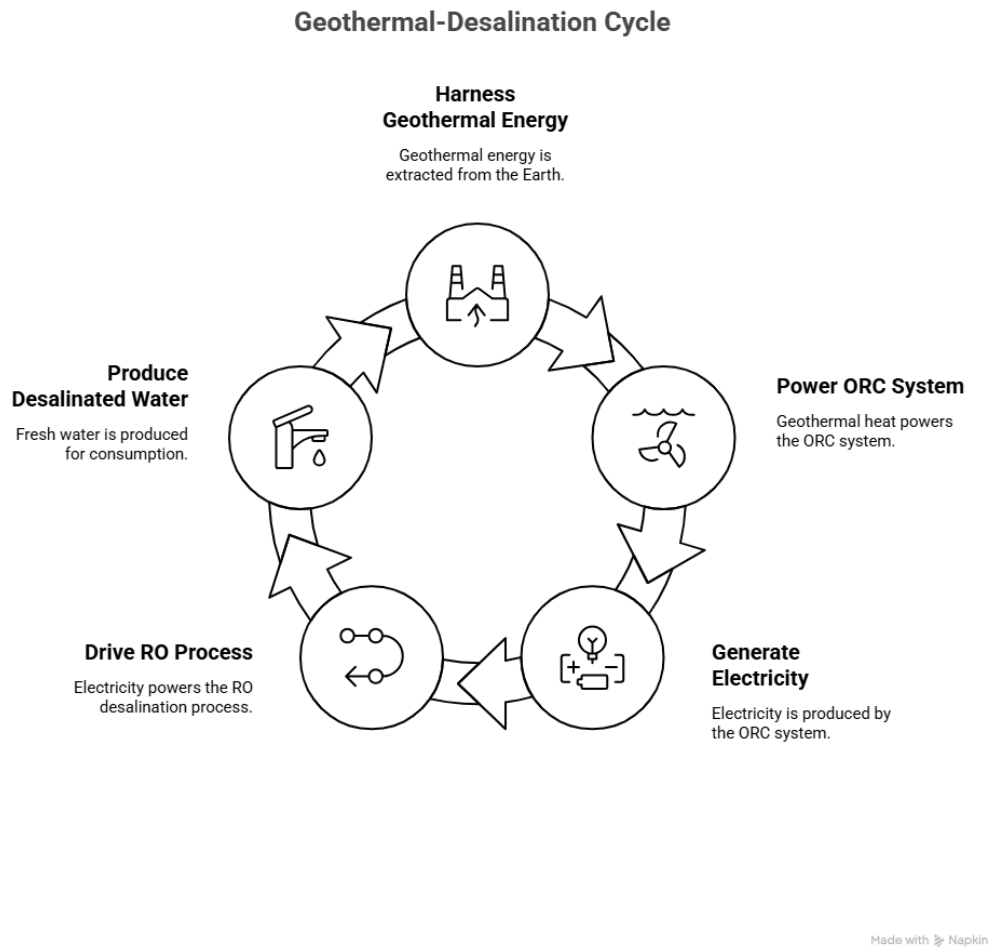


Fig. 2 Cycle depicting Reverse Osmosis in Geothermal-Desalination

1.2 Solar Energy Systems

Solar desalination leverages abundant sunlight—especially in equatorial and arid regions—to thermalize and evaporate saline water. Though still minor globally, it's expanding fast. In solar-optimized areas such as the MENA region, solar-powered RO and stills are growing [26,27]. Solar desalination can be implemented via passive thermal systems (solar stills), active thermal systems (solar-heated MSF/MED), or solar-electric systems (PV-driven RO/ED).

1.2.1 Passive solar stills

A single-basin still consists of a shallow blackened basin containing saline water, covered by a transparent sloped cover. Solar irradiation heats the water, causing evaporation; vapor condenses on the cooler underside of the cover and is collected as distillate [27]. Typical daily yields are 3–4 L/m²/day for single-slope stills, increasing to 5–6 L/m²/day for multi-slope or wick-fed designs [27,28]. This crucial method uses Solar Stills in a three-step cycle.

The first step being evaporation, where the stills consist of shallow triangular basins made up of Fiber Reinforced Plastic (FRP); the bottom of the basin is painted black to absorb solar heat effectively as shown in Fig 3. The top of the basin is covered with transparent glass tilted so that maximum solar radiation can be transmitted into the still. Solar energy from sun rays penetrates the transparent cover of the still, heating the saline water in the basin, and the freshwater evaporates, leaving behind dissolved salts and impurities. The second and third consecutively are Condensation and Collection wherein the cooler inner surface of the cover acts as a place for condensation via convection and radiation. The freshwater vapour cools down and eventually trickles down. Unlike in collection, these freshwater droplets accumulate in channels and gutters and are directed to a storage tank, where they may be further filtered if necessary to provide potable freshwater. Further enhancements increase output by ~40% at a minimum [29].

1.2.2 Active thermal systems

Solar-MSF, which via solar collectors heats feedwater to the required 90–110 °C for MSF operation, with multi-stage flashing reducing vaporization energy costs. Moreover, Solar-MED operates efficiently at 60–70 °C, making it compatible with evacuated-tube collectors or concentrating solar power (CSP) systems [30]. PV-driven RO uses Photovoltaic panels to generate electricity to power high-pressure pumps in RO plants. Specific energy consumption (SEC) for seawater RO is typically 3–5 kWh/m³, while PV array sizes are calculated based on peak insolation, plant capacity, and storage requirements [31].

1.2.3 Hybrid thermal RO

Solar energy thermally heats MSF or MED systems, merging traditional tech with solar gain. In Kuwait, solar-augmented MSF produces ~100 m³/day; Italy and the US run solar-MSF pilot plants [27]. The benefits of the usage of solar energy include zero carbon emissions, modular simplicity, and suitability for off-grid or isolated regions [27]. Although the detriments are low absolute output, unpredictable solar availability, large surface area needs, and high land footprint [27].

Solar Still Design Selection Process

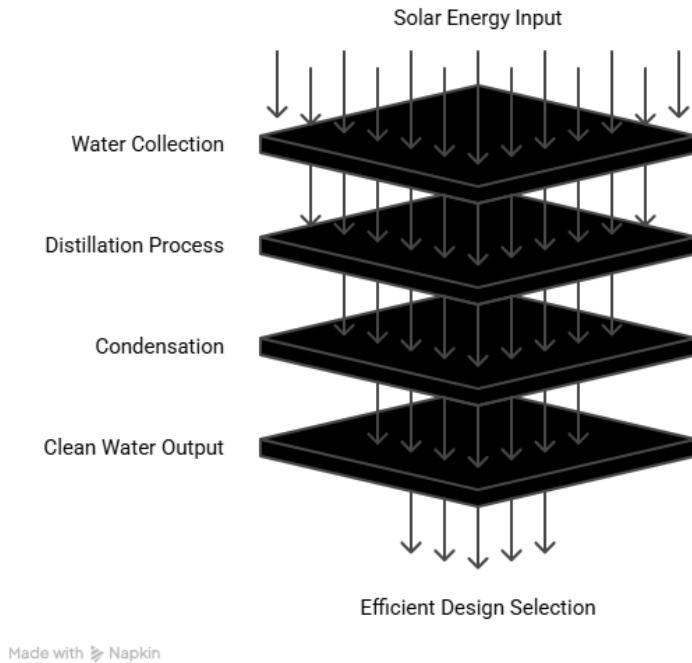


Fig. 3 Flow chart of solar still cycle; Solar still design selection process

However, needless to say, the prospects of solar energy systems are high and are believed to be on a level increase in the forthcoming years, with an expectation to see a 15% growth in this sector [32].

1.3 Wind Energy Systems

Wind energy brings a mechanical alternative to powering desalination—either via direct mechanical drive, electricity generation, or thermal hybrid systems. Over 200 wind-powered desalination plants operate globally, with combined output exceeding 1 million m³/day, particularly in the Middle East, North Africa, and the Caribbean [33]. As of late, it has established itself with over 200 wind-powered desalination plants, with many under progress, which, when summed, have an output of over 1 million m³/day. There presently exist three integration modes—Electrically, Mechanically, and Thermal hybrids—and three system types DWD, WED, and HWSD. Real-world deployments include Australia's Perth Seawater Plant, producing ~140 ML/day, with 180 GWh/year demand offset by wind energy from Emu Downs (270 GWh/year) [34,35]. Sydney's Kurnell plant (~250 ML/day, supplying 15% of

Sydney's water) is similarly backed by Capital Wind Farm energy [36]. Morocco's Agadir plant outputs 275,000 m³/day using wind and solar power [37]. On the positive side, the 'renewable' factor is especially effective in windy arid zones; lower fossil fuel dependence, and potential for cost competitiveness in such areas.

Nonetheless, the variability in wind necessitates storage or backup; high capital for turbines and integration; grid or control complexity [33]. Despite all pros and cons, with its ability to harness the power of wind, this type of renewable energy system offers a sustainable and cost-efficient solution to the world's growing water crisis.

2. Molecular Design and Advanced Materials

2.1 New Materials for the Desalination Process

2.1.1 Advanced Efficiency Nanostructured Materials

The decade's advances in nanostructured materials are transforming water desalination. Nanostructured materials such as carbon nanotubes (CNTs), graphene oxide (GO), and metal–organic frameworks (MOFs) are successfully redesigning the desalination membrane performance benchmark. Vertically aligned CNTs with diameters ≤ 1 nm form hydrogen-bonded water "wires" for near-frictionless flow at rates orders of magnitude greater than traditional membranes, and their narrow pores reject hydrated ions naturally. Molecular dynamics and laboratory-scale testing realize increases in flux by 5 $\times$ or more [34]. GO laminates offer tunable interlayer spacing to manage selective permeability oxidation and varying functional groups regulate flux and rejection of salt [35]. MOFs, with crystalline porosity engineered precisely, offer modular ion sieving and have been employed as thin-film membranes or fillers for high-performance thin-film nanocomposites (TFNs) [36]. Hybrid membranes using CNTs, GO, and MOFs within polymer matrices realize synergistic gains in mechanical strength, selectivity, and fouling resistance [37,38], with functionalizations (e.g., charged or zwitterionic moieties) further boosting ion-repelling and anti-fouling performance [39].

Therefore, in summary, Carbon nanotubes (CNTs) convey water via frictionless nanotubes; drag reduction and size selectivity enable water throughput orders of magnitude higher than conventional RO membranes, without sacrificing salt rejection when tube diameters ≤ 1 nm [34]. Their high tensile strength enables them to withstand mechanical stress at high pressure [37]. Graphene oxide (GO) membranes enable tunable interlayer spacing (0.3–1.2 nm) and functional chemistry to govern permeability and selectivity. Hydroxyl or carboxyl functionalization of GO enhances hydrophilicity at the expense of nothing but high flux and

salt rejection [38]. Metal organic frameworks (MOFs) construct ordered crystalline architectures, enabling size- and chemistry-based rejection of ions. Thin-film MOF membranes usually <100 nm possess outstanding flux/selectivity ratios [42,43,44,45]. Hybrid membranes with the introduction of added CNT, GO, MOF, and polymeric matrices provide ultra-thin, flexible, antifouling membranes beyond conventional TFC restrictions [46,47,48].

2.1.2 Biomimetic and Composite Membranes

Biomimetic membranes have been highly promising in desalination technologies as they can transport water at high flux rates with low energy requirements. The membranes can be constructed using natural or synthetic aquaporin channels, and both are highly efficient in providing high water flow rates with extremely high salt rejection selectivity [52].

Practically, Aquaporin-based membranes mimic natural water channels to achieve higher selectivity and flux. Lipid–polymer inclusions of aquaporin in polymeric matrices are long-term stable and work with optimal performance [50]. Aquaporin-mimetic artificial water channels (AWCs), on the other hand, employ synthetic backbones to mimic water conduction and avoid degradation [19,20]. Composite membranes, in addition, include nanoparticles such as TiO₂ (self-cleaning), zeolites (ion-selectivity), and graphene derivatives (thermal/chemical stability) in polymer matrices to achieve durability, high flux, and fouling resistance [21–24].

Recent research has explored the incorporation of synthetic protein channels and bioinspired nanopores into synthetic polymeric membranes as a way of mimicking highly efficient filtration systems in nature. Synthetic channels like these demonstrate high permeability with ionic selectivity maintained, thus significantly enhancing energy efficiency in reverse osmosis processes [20]. For improved mechanical strength, chemical stability, and overall desalination performance, composite membranes containing nanofillers or nanoparticles in a polymeric matrix have been developed. The addition of nanofillers like silica nanoparticles, zeolites, and TiO₂ nanoparticles provides improved antifouling behavior of the membrane and improved long-term stability, because of their exceptional chemical and thermal stability. Graphene-based composite membranes can be used in high-salinity environments [21].

2.2 Membrane Fouling Treatment

2.2.1 Surface Engineering and Antifouling Coatings

Membrane fouling remains the predominant problem in desalination, with scaling, organic fouling, and biofouling reducing membrane and operational performance. Hydrophilic

polymer brushes and zwitterionic coatings have been investigated for their anti-fouling properties because of non-adhesiveness with microbes and proteins by the formation of hydration layers [22]. Fouling lowers performance in desalination; therefore, aggressive solutions are required. Surfaces grafted with hydrophilic polymer brushes (PEG, zwitterionic) retain hydration layers that repel foulants [25]. Nano-engineered topographies lower adhesion sites for microscale fouling [26].

Responsive coatings react to environmental stimuli like pH or salinity, dynamically changing wettability or charge in an attempt to counteract changing fouling risks [27–29]

The most recent developments in dynamic surface coatings are self-adjusting layers that vary their charge density and hydrophilicity according to changing fouling conditions. These coatings dynamically modify their surface properties, hence avoiding the deposition of organic and inorganic species onto the membrane design [24].

2.2.2 Self-Cleaning and Responsive Materials

Self-cleaning materials such as TiO₂ coatings are photocatalytically active and can degrade biofilms and organic foulants under irradiation by solar light. The coatings are efficient in inhibiting biofouling and reducing chemical cleaning needs, thereby making the desalination processes green [25]. Among the recently identified families of materials are the stimuli-responsive materials that can modify their surface properties in response to a change in the environment, i.e., pH change, temperature change, or change in salinity. The materials encompass self-healing antifouling coatings, which allow for stable membrane operation over extremely long times [26].

Photocatalytic films (TiO₂, g-C₃N₄) induce light-activated reactive species to decompose fouling in situ [30,31].

Stimuli-responsive polymers contract or swell in response to heat or ionic changes to remove fouling [32].

Antimicrobial nanoparticles like Ag, Cu, and ZnO create self-sterilizing conditions without the use of toxic chemicals [33,34]. Self-healing coatings, triggered by environmental stimuli, are kept intact and retain filtration properties over time [35]. Furthermore, silver or copper nanoparticles in nanocomposite membranes have inherent antimicrobial properties that inhibit the growth of microorganisms and the formation of biofilms. The nanomaterials naturally interfere with bacterial cell membranes, thus inhibiting biofouling without using chemical disinfectants [27].

2.3 Improved Salt Exclusion and Brine Utilization

2.3.1 Salt Refining Exclusion Mechanisms

The most recent thin-film membranes use interfacial polymerization improvements to form selective layers thinner than 100 nm with controlled crosslinking—improving permeability and rejection in parallel [36]. COF and 2D nanofluidic devices use electrostatic gating to control ion migration [37–39]. Ion-selective electrode-based high-performance CDI/EDI minimizes energy expenses for industrial and brackish feeds [40–42].

2.3.2 Brine as Resource, Not Waste

Brine disposal is a major issue in the environmental impacts of desalination, with high concentrations of salts and other impurities. Emerging technologies now aim to reduce brine discharge while maximizing resource recovery. Valorization of valuable salts in brine can be achieved by the use of ion-exchange resins and hydrogel absorbents to recycle them back to industry [31]. The use of solar evaporation systems to minimize brine volume and recover water is still innovative and has minimal environmental impact. Newer methods of waste stream treatment, like electrochemical and crystallization methods, further increase the sustainability of desalination processes. For example, membrane distillation crystallization (MDC) evaporates MD brine in the process of salt harvesting, and brine is also fractionated into acids and bases by electrodialysis with bipolar membranes (EDBM) for reclamation [43,44]. Another technique is the use of Algal and bacterial bioremediation, which transforms brine impurities into biomass (biofuel or fertilizer) [45,46]. Ion-exchange and electrochemical treatment also recover valuable constituents such as lithium or magnesium for use in industry [47–49].

2.4 Partnerships Between Renewable Energy and Next-Generation Materials

To significantly lower operating costs and the environmental impact, new desalination materials can be used with renewable energy sources like solar, wind, and geothermal. Geothermal, for instance, leverages subsurface heat in volcanic/geothermal regions to drive low-energy desalination [60]. Solar thermal-responsive membranes employing solar thermal energy provide low-energy substitutes to conventional desalination processes. This is achieved by the utilization of Photothermal membranes that absorb sunlight through plasmonic particles to drive evaporation through interfacial heating [50–52]. [34]. Hybrid PV-based desalination power plants with battery backup provide a continuous and constant supply of water, while improving system ruggedness and efficiency, facilitating smart, off-grid desalination with variable solar input [53–56]. [35].

Moreover, tidal and wave power utilization for direct mechanical desalination is another off-grid and coastal solution, particularly keeping in view that they minimize electric needs and utilize durable membranes for coastal environments [36]. Development of highly integrated and energy-efficient desalination technologies that blend advanced filtration technology with renewable energy systems is a significant step towards attaining water security and

sustainability in arid zones. Advances at the interface between nanotechnology, bioengineering, and renewable energy integration are expected to render large-scale, eco-friendly desalination technology feasible.

3. Water-Energy Nexus and System Hybridization

The complex relationship between water and energy production is commonly termed the water-energy nexus. It has become a major focal point in sustainable desalination development. Conventional thermal desalination processes, which include Multi-Stage Flash (MSF) and Multi-Effect Distillation (MED), alongside membrane-based Reverse Osmosis (RO) systems, require high electricity and thermal inputs, that is why strategic renewable energy integration is the way to achieve the sustainability targets [69]. This interdependence drives the innovation in hybridization approaches that uses complementary renewable resources to ensure continuous, cost-effective freshwater production.

3.1 Renewable Energy Integration Strategies

Recent developments in hybrid renewable energy systems demonstrate remarkable potential for meeting desalination energy demands while maintaining operational reliability. A comprehensive study of PV-wind hybrid systems in Iran achieved approximately 65% electricity supply and 59% water demand fulfillment, significantly reducing grid dependence while achieving an exceptionally competitive Levelized Cost of Energy (LCOE) of 2.73¢/kWh [70]. Similarly, advanced fuzzy-optimized PV-wind-battery configurations powering RO plants in Egypt have demonstrated 100% renewable supply capability through strategic deployment of 16×100 kW wind turbines, 4127 kW photovoltaic arrays, and large-scale battery storage systems. These installations achieved LCOE values of approximately \$0.091/kWh with remarkable payback periods of ~1.1 years [71].

3.1 Key Challenges

As we study the integration of multiple renewable energy sources like solar and wind into our water desalination systems, we face a unique set of challenges. First, renewable energies are, by nature, unpredictable. Solar and wind do not produce power consistently throughout the day or year [72]. However, our desalination technologies, such as reverse osmosis or thermal units, require a steady and reliable flow of electricity to operate efficiently [73]. Sometimes, the renewable plants produce too much or too little power compared to what's needed at the moment this is called a mismatch. To fix these ups and downs, energy storage (like batteries or big water tanks) is needed to save extra energy and supply it when renewables aren't generating enough [73][74]. When multiple renewable setups work together, it's technically tricky and can be expensive since you need special inverters, control devices, and systems to make everything work together [72]. The machines that clean water (like RO membranes) require a steady flow of power. If the power changes too quickly or dramatically, it can damage the equipment. So, systems often include extra devices to smooth out energy and recover unused energy. Finally, smart controls are needed to decide when to use stored energy and how much water to produce [74], making the whole process even more complex. The summarization of this pipeline is as exhibited in Fig 4.

Fig. 4 Integrating renewable energy in Desalination Systems

3.3 Solutions

Despite the challenges of integrating renewables with desalination, there are some promising solutions and real-world success stories. For example, hybrid systems with combinations like solar PV, wind power, and battery storage have been carefully designed and optimized using advanced methods, such as fuzzy multi-criteria optimization. Take the work by Elsayed et al. (2024) they analyzed a hybrid system with solar, wind, and batteries powering a reverse osmosis plant [71]. Their optimized design met all electricity and water needs using only renewable energy, proving both technical feasibility and economic sense. [71] Looking further, case studies like that of Pietrasanta et al. (2023) showed that combining off-grid solar, wind, and water storage solutions could drive freshwater costs down to about 60 to 66 cents per cubic meter which are really impressive figures [75]. The reason hybrid PV-wind models work so well is that solar and wind naturally complement each other when solar is low, wind might be high, balancing out energy supply across day and night and through changing seasons [76].

There are more innovative approaches as well. Pairing reverse osmosis with photovoltaic-thermal collectors so called PV/T systems not only provides electricity but also preheats water for desalination, boosting system efficiency and even cooling the solar panels simultaneously [74]. Advanced control algorithms, including fuzzy logic and AI, have shown success in managing these setups, keeping everything running smoothly and preventing issues like overheating and matching variable outputs [74]. Storage also plays a vital role. Beyond batteries, pumped hydro storage has been effectively demonstrated. For instance, a standalone system combining wind power, hydro storage, and reverse osmosis was successfully operated on an island by Spyrou and Anagnostopoulos (2020). The hydro storage was used to buffer the variable wind output, ensuring steady power for desalination. Thanks to these strategies, well coordinated hybrid systems, smart controls, and effective storage we're seeing reliable and economical operation, with many projects able to achieve 60% or even up to 100% of their energy from renewables. These results prove that multi-resource hybrid systems are not just theoretically sound, but practically achievable as we move toward sustainable water solutions. [70][71][75]

References

[1] United Nations, FAO, & World Bank. (2025). Water scarcity: Two-thirds of the world's population is facing stress. *Water Scarcity Report*, 47(2), 11–15.

[2] Time. (2023). Water is the new oil in the Gulf. *Time Magazine*, 11(4), 23–25.

[3] Hadchity, M. (2025, January 5). *Saudi Arabia boosts desalinated water supply to 50% in Vision 2030 push*

[4] Wikipedia. (2025). Water scarcity: Global water stress and desalination growth. *Wikipedia*.

[5] Wikipedia. (2025). Desalination costs and environmental concerns: Cost analysis and sustainability issues. *Wikipedia*.

[6] Greco, F., Heijman, S. G. J., & Jarquin-Laguna, A. (2021). Integration of wind energy and desalination systems: A review study. *Processes*, 9(12), 2181.

[7] Wikipedia. (2025). Water scarcity indicators: Population vs. climate impact on freshwater access. *Wikipedia*.

[8] Financial Times. (2024). New climate reality stretches global freshwater supply. *Financial Times*.

[9] Sustainable Energy Research. (2015). A comprehensive decade review: Passive solar stills design & performance. *Sustainable Energy Research*.

[10] Review of efficient high productivity solar stills. (2019). *Renewable and Sustainable Energy Reviews*, 101, 197–220.

- [11] arXiv. (2020). A compact flat solar still with high performance. *arXiv*.
- [12] arXiv. (2021). Improving mass transfer & energy efficiency of solar stills through inner circulation. *arXiv*.
- [13] Financial Times. (2024). Water scarcity cost & future outlook. *Financial Times: Climate & Policy Coverage*.
- [14] Lund, J. W., & Boyd, T. L. (2016). Direct utilization of geothermal energy 2015 worldwide review. *Geothermics*, 60, 66–93.
- [15] Sener, E. (2020). Geothermal energy utilization as an alternative energy source. *Geothermal Energy*, 8(1), 1–23.
- [16] DiPippo, R. (2015). *Geothermal power plants: Principles, applications, case studies, and environmental impact* (4th ed.). Butterworth-Heinemann.
- [17] Al-Karaghoul, A., & Kazmerski, L. L. (2018). Energy consumption and water production cost of conventional and renewable-energy-powered desalination processes. *Renewable & Sustainable Energy Reviews*, 67, 1267–1275.
- [18] El-Dessouky, H. T., & Ettouney, H. M. (2002). *Fundamentals of saltwater desalination*. Elsevier Science.
- [19] Tiwari, G. N., & Tiwari, A. K. (2019). Research and development of solar still: A review of enhanced desalination systems. *Applied Energy*, 236, 555–584.
- [20] Lattemann, S., & Höpner, T. (2008). Environmental impact and impact assessment of seawater desalination. *Desalination*, 220(1–3), 1–15.

[21] Banat, F., Jwaied, N., Zayed, Y., & Rommel, M. (2007). Solar thermal desalination technologies. *Desalination*, 203(1–3), 228–235.

[22] Sutaryo, S., & Daud, W. R. W. (2018). Organic Rankine Cycle (ORC) system implementation using geothermal energy for power generation. *IOP Conference Series: Materials Science and Engineering*, 335(1), 012044.

[23] Ali, O. M., & Hilal, N. (2008). Adsorptive desalination: Potential of the current generation of adsorbents and future perspectives. *Desalination*, 232(1–3), 90–101.

[24] Hamed, M. B., Kripalani, D., & Dincer, I. (2013). A review of thermal energy storage in the desalination industry. *Renewable Energy*, 66, 514–527.

[25] Al-Shammiri, A. A., Klein, S. A., & Krarti, M. (2008). Economic analysis of two geothermal desalination systems. *Desalination*, 456, 61–70.

[26] Abdelkareem, M. A., Hamdan, M. O., & Sopian, K. (2020). Recent advancements in hybrid renewable energy systems supporting desalination. *Desalination*, 489, 114506.

[27] Chaibi, M. T. (2000). Solar desalination for domestic applications: A review. *Desalination*, 127(2–3), 119–133.

[28] Kalogirou, S. A., & Zhang, X. (2004). Experimental investigation of a single-basin solar still with internal and external reflectors. *Desalination*, 171(2), 187–202.